

BRAC UNIVERSITY
Department of Computer Science and Engineering

CSE 360: Computer Interfacing

Assignment

Full Marks: 20

Answer All 3 following Questions.

[Figures in the right margin indicate marks]

CO1	Question 1. a) USB type-C is most widely used now - explain why. b) Write 3 comparisons between serial transmission and parallel transmission.	[2] [1]
CO2	Question 2. Mary is a hardware engineer and she is working on a project that requires communication between a microcontroller and peripheral devices. He has two options for communication protocols: the first one is very simple but supports multi-master and multi-slave configurations, while the second is widely used but does not support multiple masters. a) What is the name of the second protocol? Explain step by step how the communication of the second protocol takes place, with a proper diagram. b) Suppose you want to send - "10101010", "10001111", "001101101" using the second protocol and now draw the necessary data packet for it. c) Write a comparison between the first and second communication protocol mentioned in the scenario.	[3] [3] [2]
CO3	Question 3. a) Draw the simplified system architecture of STM32 microcontroller. b) List down the registers that are required for the GPIO operations in STM32. Harvey is making a project with STM32F446RE and a few devices. He has decided to work with a few digital ports of the STM32. Now, he is testing the pins of the port C of STM32. c) Write down his commands for following configuration: i) Activate the port C of STM32; ii) Pin number - 7 of this port will be used for output where data transfer rate is high speed open drain iii) Pin number - 9 of the same port will be used for input where the connection will be used as Pull- up d) After configuring the STM32 following configuration which port/ports will be activated? Configuration: RCC → AHB1EN = (1<<3); RCC → AHB1EN &= ~(0<<5); RCC → AHB1EN = (1<<1) = (1<<0);	[2] [2] [3] [2]